

META-ANALYSIS

Nephrology

Effect of strict intraoperative blood pressure management strategy on postoperative acute kidney injury in non-cardiac surgery: A meta-analysis of randomised controlled trials

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Abstract

Background: Acute kidney injury (AKI) is one of the most serious perioperative complications. 20% to 40% of high-risk patients who undergo non-cardiac surgery have AKI and those with AKI are eight-times more likely to die within 30 days after surgery. It may be related to intraoperative hypotension, which is mainly caused by vasodilatory and cardiodepressant effects of anaesthesia, and/or hypovolemia. Strict intraoperative blood pressure management strategy (strict BP management) is a potential option to prevent postoperative AKI. This strategy refers to additional administration of vasoactive agents under the premise of a protocolised fluid delivery. The efficacy of strict BP management for preventing postoperative AKI in non-cardiac surgery patients was assessed by a meta-analysis.

Methods: We systematically retrieved randomised controlled trials (RCTs) and compared strict BP management with conventional therapy control on effect of postoperative AKI in non-cardiac surgery patients, which were published on PubMed, EMBASE, Cochrane library and Web of Science databases before October 5, 2020. Ultimately, a meta-analysis of all RCTs eligible for inclusion criteria was performed.

Results: Five RCTs, comprising 1485 patients, were included in the meta-analysis. Strict BP management was associated with a reduced incidence of postoperative AKI [relative risk (RR) = 0.73, 95% confidence interval (CI): 0.58-0.92, $P = .007$]. No significant difference was found between strict BP management group and conventional therapy control in mortality at longest follow-up available (RR = 0.92, 95% CI: 0.68-1.25, $P = .60$). In the subset analysis, the studies using supranormal BP management target was significantly lower in the incidence of postoperative AKI (RR = 0.65, 95% CI: 0.51-0.82, $P = .0003$).

Conclusion: Strict BP management is significantly more effective than conventional therapy for the prevention of postoperative AKI. Supranormal target of intraoperative BP management may be considered a more appealing option for the prevention of AKI.

1 | INTRODUCTION

Acute kidney injury (AKI) is one of the most serious perioperative complications with a high rate of morbidity and mortality.¹⁻³ According to Bauerle et al,⁴ AKI occurs in 20% to 40% of high risk patients who undergo non-cardiac surgery. Kheterpal et al⁵ showed that surgical patients with postoperative AKI are eight-times more likely to die within 30 days after surgery. However, there is no effective treatment strategy for AKI in clinical care currently.^{6,7} Thus, finding a preventing strategy is the key to improving the prognosis of perioperative AKI patients.

Intraoperative hypotension is an important risk factor for postoperative AKI after non-cardiac surgery.^{8,9} It may induce decreasing in renal perfusion, renal tissue ischemia and hypoxia, and finally, AKI.¹ It is reported that AKI might occur when mean arterial pressure (MAP) < 65 mm Hg. This risk increases when blood pressure (BP) becomes progressively lower or duration of intraoperative hypotension becomes longer.^{8,10-13} However, up to 40% of cases demonstrate a MAP less than 65 mm Hg for at least 10 to 12 min among patients undergoing non-cardiac surgery.⁹

Intraoperative hypotension is a frequent consequence of hypovolemia, as well as the vasodilatory and cardiodepressant effects of anaesthesia.¹ For these reasons, intravenous fluids and/or vasopressors can be used to prevent the occurrence of intraoperative hypotension and protect renal function.¹⁴ However, there's no consensus on the minimal acceptable BP threshold that anaesthetised patients should be maintained to prevent AKI. Clinical trials have examined strict intraoperative BP management strategy (strict BP management) as a potential option to prevent postoperative AKI.¹⁴ This strategy refers to a more strict BP management threshold with additional administration of vasoactive agents, under the premise of a protocolised fluid delivery. However, no meta-analysis has evaluated the efficacy of this strategy for preventing postoperative AKI.

Based on the previous considerations, we performed a meta-analytic study about the effect of strict BP management on postoperative AKI in non-cardiac surgery patients compared with conventional therapy. We hypothesised that strict BP management would protect renal function, and therefore, reduce the incidence of postoperative AKI.

What's known

- Systematic searches were performed in PubMed, EMBASE, Cochrane library and Web of Science databases for relevant trials published before October 5, 2020.
- Trials that compared strict intraoperative blood pressure management strategy with conventional therapy control on effect of postoperative acute kidney injury in non-cardiac surgery patients were included in the meta-analysis.

What's new

- In non-cardiac surgery patients, strict intraoperative blood pressure management strategy, which maintains normal or supranormal blood pressure by vasopressors under the premise of a protocolised fluid delivery, is more effective than conventional therapy for the prevention of postoperative acute kidney injury.

2 | METHODS

The conduct and reporting of this meta-analysis followed the Cochrane Collaboration methodology¹⁵ and PRISMA statement.¹⁶ The protocol is registered at PROSPERO (registration number: CRD42020178340, available at <https://www.crd.york.ac.uk/prosp/ero/>)

2.1 | Search strategy

Two investigators (M.T. and J.L.) independently searched PubMed, EMBASE, Cochrane library and Web of Science databases from the inception dates to October 5, 2020 for eligible studies. Besides, records from other resources were also identified. We hand-scanned references of retrieved articles and pertinent reviews to identify other eligible trials (backward snowballing). Abstracts from recent

Database	Search strategy
Web of Science	#1 TS=("blood pressure") OR TS=("arterial pressure") OR TS=(hypertension) OR TS=(hypotension) OR TS=("systolic pressure") OR TS=("diastolic pressure") OR TS=("mean arterial pressure") OR TS=(MAP) OR TS=("arterial pressures") OR TS=("artery pressure") OR TS=(hypertensive) OR TS=(hypotensive) #2 TS=(intraoperative) OR TS=(perioperative) OR TS=(intra-operative) OR TS=(intraoperatively) OR TS=(perioperatively) #3 TS=("acute kidney injury") OR TS=("acute kidney failure") OR TS=("acute renal failure") OR TS=("acute renal injury") OR TS=("acute kidney insufficiency") OR TS=("acute renal insufficiency") OR TS=("acute tubular necrosis") OR TS=(ARI) OR TS=(AKI) OR TS=(ARF) OR TS=(AKF) OR TS=(ATN) #4 #3 AND #2 AND #1

TABLE 1 Search strategy in Web of Science database

international conferences were searched for additional studies. Google Scholar was also searched for potentially relevant articles that cited eligible trials. We contacted authors to clarify eligibility or obtain relevant unpublished data. No language or publication status restrictions were imposed. Publication status means studies from a range of status, including unpublished studies, partially published studies, studies published in "grey" literature sources and studies published in peer-reviewed journals. The search strategy in Web of Science database is shown in Table 1 and others are shown in Table S1.

2.2 | Study selection

Studies were considered eligible if they met the following PICOS criteria: (i) Population: adults (age 18 years or over) undergoing non-cardiac surgery; (ii) Intervention: strict BP management. The strict BP management was defined as intraoperative monitoring and manipulation of BP parameters to reach normal ($\text{MAP} \geq 70$ mm Hg/ MAP decrease less than 30% from the baseline¹⁷) or supranormal (the BP management threshold higher than normal) values by fluids and/or vasoactive therapy; (iii) Comparison intervention: conventional therapy; (iv) Outcome: the outcome of interest was postoperative AKI; and (v) Study design: the study design was a randomised controlled trial. Exclusion criteria were: (i) studies on organ transplantation or emergency procedures for ruptured vessels; (ii) studies with therapy titrated to the same BP management threshold in both groups; (iii) studies published in retracted papers.

2.3 | Quality assessment

Two investigators (M.T. and J.L.) independently assessed the methodological quality of each included trial according to the Cochrane risk-of-bias criteria,¹⁵ including generation of random sequence generation, allocation concealment, blinding of participants and personnel, blinding of outcome assessment, completeness of outcome data, selective reporting and other bias. The other bias domain included the classic items reported by the "Cochrane Handbook for Systematic Reviews of Interventions".¹⁵ Each quality item was graded as low risk, high risk or unclear risk. Owing to the nature of the intervention, blinding of personnel seemed difficult and was therefore not judged as crucial for bias assessment. The included trials were graded as low quality, high quality or moderate quality based on the following criteria: (i) trials were considered low quality if either randomisation or allocation concealment was assessed as a high risk of bias, regardless of the risk of other items; (ii) trials were considered high quality when both randomisation and allocation concealment were assessed as a low risk of bias, and all other items were assessed as low or unclear risk of bias in a trial; (iii) trials were considered moderate quality if they did not meet criteria for high or low risk.¹⁸

2.4 | Data extraction

Data were independently collected by two investigators (M.T. and J.L.), with any discrepancy resolved by reinspection of the original article and consensus with each other. Data abstraction included general information (first author, publication year, country, sample size), patients characteristics (age, gender, ASA physical status, comorbidity), procedure characteristics (type of anaesthesia, length of surgery, elective or emergent surgery, type of surgery), intraoperative BP management strategy (timing, threshold type and threshold, therapeutic intervention) and details on AKI (definition, incidence).

The primary outcome was AKI, whichever definition was used, and secondary outcome was mortality at the longest follow-up available.

2.5 | Statistical analysis

We performed meta-analysis to calculate risk ratios (RRs) and 95% confidence intervals (CIs) using the Mantel-Haenszel statistical method. A random-effects model was used to pool the data. Statistical heterogeneity and inconsistency were assessed by using the Q and I^2 tests, respectively. When the Q test P value was $<.10$ and/or the I^2 was $> 50\%$, heterogeneity and inconsistency were considered significant.

We planned several subgroup analyses for the main outcome on the basis of the following: (i) quality of trials; (ii) timing of BP management; (iii) modality of BP management (iv) threshold of BP management; (v) definition of AKI. Subgroup analyses were conducted only if there were at least two studies in each subgroup. If this meta-analysis include no more than two of above features used for grouping and each has only one pair of subgroup means to compare, then the P value was set at .05, otherwise an adjusted P value of .01 was performed for subgroup analysis.

Sensitivity analyses were performed by analysing the data with a fixed or random effects model and using other summary statistics. Because the number of studies for each outcome was smaller than 10, publication bias was not examined.¹⁵ Statistical significance was set at $P = .05$. The meta-analysis was performed using Review Manager (RevMan, version 5.3; The Nordic Cochrane Center, The Cochrane Collaboration, Copenhagen, Denmark, 2014).

3 | RESULTS

3.1 | Search results and study characteristics

The search strategy identified 2320 citations and, after exclusion of inadequate reports (Supporting Information for records excluded), five trials were included in the meta-analysis (Figure 1).^{14,19–22}

The characteristics of the included studies, intervention details of analysed studies and renal outcome definition are shown in Tables 2 and 3. The included studies involved a grand total of 1485

patients. All included articles evaluated the effect of strict BP management on postoperative AKI as primary or secondary outcome and had a population sample of adult patients, undergoing both elective or emergent surgical procedures. The studies were performed in United Kingdom, France, Germany and China from 2016 to 2019 and were all published in English.

3.2 | Quality assessment

Among included studies, five (100%) adequately generated their randomisation sequence;^{14,19-22} three (60%) adequately concealed allocation;^{14,19,20} two (40%) blinded participants,^{14,20} four (80%) blinded outcome assessors;^{14,19,20,22} five (100%) registered study protocols at clinical trial registries and did not selectively report outcomes.^{14,19-22} No trial had missing outcome data >20%. Three trials (60%) were free from conflicts of interest or sponsorship bias (Figure 2).²⁰⁻²²

3.3 | Strict intraoperative blood pressure management strategy vs conventional therapy control

3.3.1 | Incidence of postoperative AKI

Among the 1485 patients in the included studies, 334 developed postoperative AKI. Of these, 149 belonged to strict BP management group (17.9%), and 185 to conventional therapy control group

(28.4%). Strict BP management was associated with a reduced incidence of postoperative AKI (RR = 0.73, 95% CI: 0.58-0.92, $P = .007$; $I^2 = 31\%$, $P_{\text{heterogeneity}} = .21$; Figure 3).

3.3.2 | Incidence of mortality

No significant difference was found with strict BP management group vs conventional therapy control in mortality at longest follow-up available (RR = 0.92, 95% CI: 0.68-1.25, $P = .60$; $I^2 = 0\%$, $P_{\text{heterogeneity}} = .86$; Figure 4). Three trials that involved 1178 participants provided data on mortality at 28 to 30 days.^{14,19,22} No significant difference in mortality at 28 to 30 days was found (RR = 1.10, 95% CI: 0.67-1.83, $P = .70$; $I^2 = 0\%$, $P_{\text{heterogeneity}} = 1.0$; Figure S1).

3.4 | Subgroup analysis and sensitivity analysis

The incidence of postoperative AKI was significantly lower in "high-quality" trials (RR = 0.65, 95% CI: 0.42-0.99, $P = .04$; $I^2 = 31\%$, $P_{\text{heterogeneity}} = .23$; Table S2), but no difference was observed in "non-high-quality" trials (RR = 0.83, 95% CI: 0.66-1.04, $P = .10$, $I^2 = 0\%$, $P_{\text{heterogeneity}} = .43$; Table S2). The subset analysis including studies using supranormal BP management threshold showed an RR of 0.65 (95% CI: 0.51-0.82). BP management to maintain normal values yielded no significant benefits (RR = 0.79, 95% CI: 0.50-1.25) (Table S2). No significant subgroup differences were found by quality of trials (high-quality vs non-high-quality) and threshold of BP

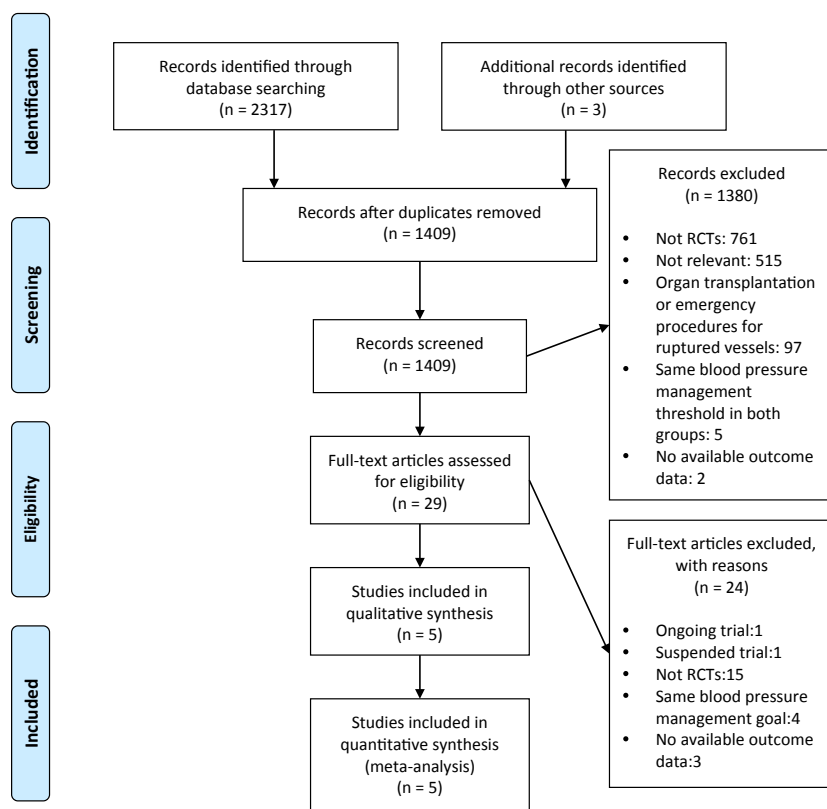


FIGURE 1 Flow diagram of study selection

TABLE 2 Patient characteristics of the included trials

Trials	Patient characteristics			Comorbidity (%)					Procedure characteristics (%)				
	First author, year (country)	Total patients randomised (completed)	Mean age	Gender	ASA	Stroke	HT	DM	CAD	Renal disease	General Anaesthesia	Length of surgery	Emergency surgery
n													
Davies, 2019 (United Kingdom)	241 (240)	86.5	58.1	I-IV	18.3	53.9	14.5	24.5	13.3	55.6 ^a	67.1 (25.3)	100.0	Femoral fracture (100.0)
Futier, 2017 (France)	298 (292)	69.8	15.1	II-IV	-	82.2	51.4	17.8	15.4	100.0	275.2 (138.9)	15.4	Abdominal (100.0)
Schmid, 2016 (Germany)	193 (180)	66.0	23.3	I-III	-	56.1	20.0	17.2	5.0	100.0	>180.0	-	Oesophageal (75.6)
Schmid, 2019 (Germany)	132 (127)	80.6	70.1	II-III	-	-	21.3	14.2	27.6	100.0	86.6 (37.3)	100.0	Hip fracture (100.0)
Wu, 2017 (China)	678 (646)	73.3	34.4	I-III	3.6	100	24.1	-	0	100.0	>60.0	0	Major gastrointestinal surgery (100.0)

Abbreviations: -, not available; ASA, American Society of Anaesthesiologists-Physical Status; CAD, coronary artery disease; DM, diabetes mellitus; HT, hypertension; Mean (SD), mean (standard deviation).

^aThe remaining patients received spinal anaesthesia. Both parts of patients were included in the meta-analysis.

TABLE 3 Intervention details of analysed studies and renal outcome definition

Trials	Timing of management	Threshold of management	Modality of management	Definition of acute kidney injury
Davies, 2019	Intraop	70%-130% baseline MAP	Fluids, phenylephrine, metaraminol	30% increase of baseline creatinine or/and <0.5 ml/h diuresis
Futier, 2017	Intraop up to 4 hours postop	90%-110% baseline SBP	Norepinephrine, ephedrine	RIFLE
Schmid, 2016	Intraop up to 72 hours postop	MAP > 70 mm Hg	Fluids, norepinephrine, dobutamine	RIFLE
Schmid, 2019	Preop from PACU admission before surgery for PACU postop	MAP > 70 mm Hg	Fluids, catecholamines	AKIN
Wu, 2017	Intraop	MAP 65-79 mm Hg; MAP 80-95 mm Hg; MAP 96-110 mm Hg	Fluids, norepinephrine, phenylephrine, nitroglycerin, phentolamine, esmolol, atropine	KDIGO

Abbreviations: AKIN, Acute Kidney Injury Network; Intraop, intraoperative; KDIGO, Kidney Disease: Improving Global Outcome; MAP, mean arterial pressure, the average arterial pressure throughout one cardiac cycle, systole and diastole; PACU, postanesthesia care unit; Postop, postoperative; Preop, preoperative; RIFLE, risk, injury, failure, loss, and end-stage kidney injury; SBP, systolic blood pressure.

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias)	Blinding of outcome assessment (detection bias)	Incomplete outcome data (attrition bias)	Selective reporting (reporting bias)	Other bias
Davies 2019	+	+	?	+	+	+	?
Futier 2017	+	+	+	+	+	+	?
Schmid 2016	+	?	?	?	+	+	+
Schmid 2019	+	+	+	+	+	+	+
Wu 2017	+	?	?	+	+	+	+

FIGURE 2 Risk of bias summary

management (supranormal vs normal); test for subgroup differences *P* values were .32 and .45, respectively. We were unable to perform subgroup analysis for timing of BP management, modality of BP management and definition of AKI as there was less than two studies in one given subgroup.

Sensitivity analysis demonstrated the robustness of results of postoperative AKI and mortality using a fixed or random effects model and using other summary statistics (Table S3).

4 | DISCUSSION

The main result of the present meta-analysis shows that the incidence of postoperative AKI is significantly reduced by strict BP management. For mortality at longest follow-up available and mortality at 28 to 30-days, no significant difference was found with strict BP management vs conventional therapy control. These findings were robust throughout sensitivity analyses, thereby confirming that strict BP management strategy prevents postoperative AKI.

Recently, several studies reported that the intraoperative hypotension plays an important role in affecting renal perfusion, which is one of the key pathogenetic events in promoting postoperative AKI.^{8,23} Protection against intraoperative hypotension mainly relies on maintaining adequate intravascular volume and/or vasopressors. The present study suggests that a "strict" intraoperative BP control using vasopressors, with a protocolised haemodynamic algorithm to guide fluid delivery and to maximise stroke volume, could reduce the incidence of AKI.

For the Strict BP management, an interesting point of debate is how much the minimal acceptable BP should be maintained in anaesthetised patients. Some studies have set up BP targets to supranormal values (systolic BP within 10% of the baseline; MAP 80-95

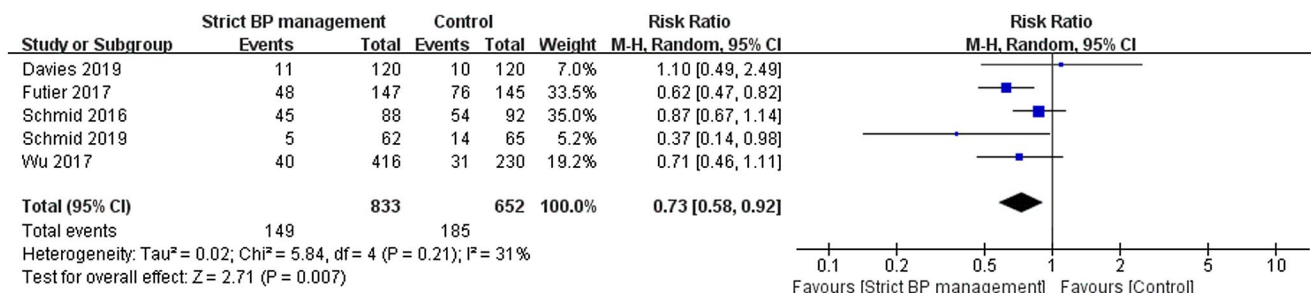


FIGURE 3 A forest plot showing the incidence of postoperative acute kidney injury in patients who received strict intraoperative blood pressure management strategy or conventional therapy controls. Strict BP management, strict intraoperative blood pressure management strategy group; Control, conventional therapy control group

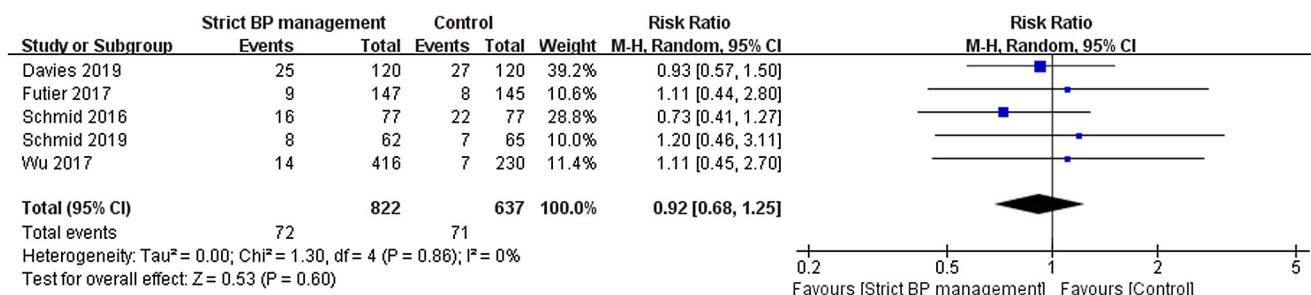


FIGURE 4 A forest plot showing the incidence of mortality at longest follow-up available in patients who received strict intraoperative blood pressure management strategy or conventional therapy controls. Strict BP management, strict intraoperative blood pressure management strategy group; Control, conventional therapy control group

mm Hg, MAP 96–110 mm Hg), as proposed by futier et al¹⁴ and Wu et al,²² whereas others to normal values (MAP $\geq$ 70 mm Hg/MAP decrease less than 30% from the baseline).^{19–21} In the subgroup of supranormal targets, strict BP management showed a larger effect (risk ratio 0.65) for decreasing the probability of postoperative AKI compared with the control, thus could protect renal function. The reason might be that a supranormal intraoperative BP management target will increase renal perfusion and oxygen delivery, and then reduce postoperative AKI, with the premise of a protocolised haemodynamic algorithm to guide fluid delivery and maximise stroke volume. In the subgroup of normal targets, although no difference in postoperative AKI rate was observed, strict BP management group tended to reduce the incidence of postoperative AKI compared with the control. If more participants are included in the subgroup of normal targets, the differences might be detected on postoperative AKI rate. Both targets, especially the supranormal one, usually achieved through the use of vasoactive agents, such as norepinephrine, phenylephrine. This result suggests that strict BP management (especially a supranormal intraoperative BP management target) with the use of vasoactive agents will be of valuable clinical implications.

To our knowledge, this is the first meta-analysis on the target of intraoperative BP management strategy and its impact on postoperative AKI. Previous trials have suggested a lower incidence of organ dysfunction with goal-directed haemodynamic optimisation therapy (GDT) during surgery. The targets used in the GDT protocol include indices of preload responsiveness, cardiac output (CO) or oxygen delivery (DO_2), or other indirect indices of DO_2 , such as lactate and

venous saturation. However, the BP management target was seldom included or different from one and another in the GDT protocol.²⁴ The present study has valuable clinical implications that a strict BP management would be of good benefits for postoperative AKI. Probably because, on one side patients with a reduced physiological reserve may benefit from strict BP management to increase oxygen delivery and counteract renal hypoperfusion; on the other side, fluid administration in clinic might not be adequate since the lack of wide use of invasive or non-invasive haemodynamic monitoring, in which condition proper use of vasoactive agents drugs to increase BP may increase organ perfusion pressure, thus bring additional benefits.

This review was not without limitations. First, although an exhaustive literature search was performed, only a small number of studies was found. Fortunately, the quality of the trials included in our meta-analysis was high (three trials were awarded high-quality and two moderate-quality), and each of the included trials recruited quite a number of patients. Second, no attempt was made to correct the type or quantity of vasoactive agents given, because they are inconsistently reported in the literature and have a demonstrable wide variability in their dosing across studies. However, the results remain consistent across a number of subgroups and sensitivity analyses. Third, the mean ages of the patients in these trials are all above 65 years old, the results should be cautiously extrapolated to other patients. More well-structured RCTs taking into account both preoperative blood management and renal function, complexity of surgery, target of intraoperative blood management, with protocolised approach to fluid administration are needed to clarify this issue.

In conclusion, the results of our study suggest that strict BP management is significantly more effective than conventional therapy for the prevention of postoperative AKI. Available evidence suggests that supranormal targets of intraoperative BP management may be considered a more appealing option for the prevention of AKI.

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DISCLOSURES

The authors report no conflict of interest.

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SUPPORTING INFORMATION

Additional Supporting Information may be found online in the Supporting Information section.

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